The Clebsch orientation cubic: arithmetic covers and icosahedral harmonics

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Abstract

The Clebsch matching geometry carries a two-valued orientation whose first odd observable is cubic. We relate this finite orientation to Hitchin's ordered-icosahedron double cover. The paper's central target identifies the golden Clebsch torsor with the fibre of the quadratic cover $w^2 = 5J_0$, and identifies its reduction modulo 11 with the Clebsch sheet exchange. An exact tensor calculation already shows that the signed matching cubic and the Clebsch cubic span the same invariant line over $\mathbf{F}_{11}$. The remaining arithmetic specialization and integral comparison are isolated as explicit certification gates. In characteristic zero, the same cubic is the restriction of the standard $\ell = 6$ Gaunt cubic to the four-dimensional icosahedral face-axis channel. A further target asks whether the same real quadratic field is recovered from the relative position of the two icosahedral multiplicity structures in the intermediate Jacobian of the Klein cubic.

Keywords. Clebsch configuration; Hitchin cover; icosahedral invariant; arithmetic double cover; Klein cubic; intermediate Jacobian.

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1 Introduction

The Clebsch matching configuration has two sheets. Forgetting the sheet retains the unordered matching data; recovering it requires an odd observable. Paper II constructs such an observable over $\mathbf{F}_{11}$: a signed cubic tensor on a ten-dimensional matching quotient. The question here is arithmetic. Does this finite orientation arise as a specialization of a characteristic-zero double cover?

Hitchin's ordered-icosahedron construction supplies a candidate. In an invariant chart its quadratic cover has the form

$$w^2 = 5J_0.$$

On the Clebsch four-space the invariant J_0 is expected to restrict to $16\sigma_3^2$. Away from $\sigma_3 = 0$, division by $4\sigma_3$ then leaves the constant quadratic torsor

$$\operatorname{Spec} \mathbf{Q}(\sqrt{5}) \longrightarrow \operatorname{Spec} \mathbf{Q}.$$

The factor 5 records the orientation field; the cubic records the first place at which orientation becomes visible.

Principal target. The central theorem will identify the golden Clebsch orientation torsor with this Hitchin fibre and prove that its deck exchange reduces modulo 11 to the finite sheet involution T_{11}. This statement is partitioned in the trust ledger as `HC-1`, `HC-2`, `HC-3`, and `HC-4`. The tensor clause `HC-3` is certified; the integral and specialization clauses remain gated.

Harmonics. The axes through opposite icosahedral faces give a canonical ten-dimensional subspace of the degree-six spherical harmonics. Its four-dimensional summand is the Clebsch module, and the standard Gaunt cubic restricts there to a nonzero multiple of σ_3. This identifies the orientation observable with a symmetry-resolved Steinhardt-type cubic, not merely with an analogous invariant.

Strong-form target. The intermediate Jacobian of the Klein cubic has no invariant elliptic factor. Its restriction to either icosahedral subgroup nevertheless has a canonical two-dimensional multiplicity Hodge structure. We ask whether the relative position of the two such structures recovers $\mathbf{Q}(\sqrt{5})$. A positive answer would make this a second structural theorem, linking the finite orientation field to a characteristic-zero period invariant. Section 6 is deliberately detachable: a negative calculation removes the elevation without changing the principal theorem.

Proof architecture. Section 2 separates the general quadratic-cover mechanism from its invariant-theoretic input. Section 3 proves the finite cubic bridge. Section 4 isolates the exact integral specialization still required. Section 5 constructs the harmonic channel, and Section 6 develops the stronger Klein target. The final sections admit only consequences of these theorems and state the proof modes used.

2 Orientation as a quadratic cover

Let R be a domain with an involution ι. Write

$$R^+ = \{r : \iota(r) = r\}, \qquad R^- = \{r : \iota(r) = -r\}.$$

The elementary algebra behind the construction is that one nonzero odd element trivializes the odd summand after localization.

Lemma 2.1 (Localized odd generator)**.** `ALG-1`*. Assume that* 2 *is invertible in* R*, and let* $C \in R^-$ *be nonzero. After localizing at* C^2*,*

$$R[(C^2)^{-1}] = R^+[(C^2)^{-1}] \oplus C\, R^+[(C^2)^{-1}].$$

Proof. Every $r \in R$ has the decomposition

$$r = \frac{r + \iota(r)}{2} + \frac{r - \iota(r)}{2}.$$

The first term is invariant. If r is odd, then r/C is invariant after C is inverted, because both numerator and denominator change sign. Since inverting C^2 also makes C invertible, the odd summand is C times the invariant summand. The intersection is zero because an element in it is both invariant and odd. $\square$

Thus an orientation-forgetting quotient is generically quadratic whenever its odd algebra is generated by one element. For the sign-twisted icosahedral action on the Clebsch four-space, the first candidate odd generator is the cubic

$$\sigma_3(y) = \frac{1}{3}\sum_{i=1}^{5} y_i^3, \qquad \sum_{i=1}^{5} y_i = 0.$$

The invariant-ring identification and its characteristic boundary are the classical inputs assigned to `HC-1`.

Target cover. On the open Clebsch chart where $\sigma_3 \neq 0$, the ordered-icosahedron cover should pull back to

$$w^2 = 5J_0, \qquad J_0 = 16\sigma_3^2.$$

Equivalently, $w/(4\sigma_3)$ squares to 5. This is the exact point at which the geometric cover becomes the golden constant torsor. We do not promote this paragraph to a theorem until C652 and C653 close the normalization, integral-model, and citation gates.

3 The Clebsch cubic bridge

Let G be the order-60 stabilizer of the H_3-invariant matching in $\mathrm{PGL}_2(\mathbf{F}_{11})$, and let Ω be its 22-point matching orbit. For a reference matching M_0, put

$$W = \langle q_M - q_{M_0} : M \in \Omega \rangle .$$

The two $\mathrm{PSL}_2(\mathbf{F}_{11})$-sheets define signs $\epsilon(M) \in \{\pm 1\}$ and the symmetric tensor

$$T = \sum_{M \in \Omega} \epsilon(M)[q_M - q_{M_0}]^{\otimes 3} \in \mathrm{Sym}^3(W).$$

Theorem 3.1 (Finite tensor bridge)**.** *There is a G-equivariant isomorphism*

$$S : \mathbf{F}_{11}^{\binom{[5]}{2}} \longrightarrow W$$

such that, after using the standard pairing to identify the permutation module with its dual and restricting along

$$E(y)_{\{i,j\}} = y_i + y_j, \qquad \sum_i y_i = 0,$$

one has

$$E^*(S^{-1})^{\otimes 3} T = 4\sigma_3.$$

In particular, the signed matching cubic and the Clebsch cubic span the same nonzero invariant line over $\mathbf{F}_{11}$.

Proof. Conjugation on the normalizers of the five Klein four subgroups identifies G with A_5 in its degree-five action. Hence it acts on the ten two-subsets. The matching construction gives a second ten-dimensional representation on W. Solving $\rho_W(g)S = S\rho_P(g)$ produces an invertible intertwiner, and the recorded matrix satisfies this identity for all 60 elements.

Transporting the 220 symmetric coordinates of T through this intertwiner and contracting with E gives a polarized 4^3-tensor whose entries are 0 when all indices agree and 6 otherwise. The polarization of σ_3 has entries 0 and $-1/3 = 7$, respectively. Since $4 \cdot 7 = 6$ in $\mathbf{F}_{11}$, the two tensors agree. A primary exact calculation and an independent reconstruction verify the representation, intertwiner, and all 64 contracted entries. The final contraction is also checked in Lean. $\square$

Remark 3.2 (Normalization). The characteristic-zero Gaunt scalar presently available has denominator divisible by 11. The theorem therefore identifies a common integral cubic line; it is not obtained by reducing that rational scalar modulo 11.

4 Arithmetic specialization

The remaining core argument has three interfaces. Each must be proved with an explicit integral normalization.

1. The two golden fibres must be identified with the two icosahedral parents A_5^- and A_5^+, including the comparison matrices and their common A_4.
2. The order-four exchanger must be computed together with its spinor square class and its reduction modulo 11.
3. The reduced exchanger must agree with the finite deck involution T_{11}, not merely have the same order or interchange the same two unlabelled sets.

The same distinction applies to the Mathieu formulation. Two M_{11} parents, their $\mathrm{PSL}_2(11)$ intersection, and Hadamard row–column exchange give a marked-torsor corollary only after their carriers have been identified with the reduced Hitchin construction. Group orders and abstract conjugacy classes do not supply this identification. This conditional claim is `HC-5`.

Integral boundary. The abstract quadratic cover gives, on its nonbranch locus,

$$\#X_f(\mathbf{F}_q) = 1 + \chi_q(5J(f)).$$

Its geometric interpretation over finite fields currently holds only away from an unspecified finite set of primes. The final theorem will name the largest justified integral base supplied by an explicit model; it will not replace that finite exceptional set by $p \neq 2, 5$ without proof.

5 The degree-six harmonic realization

Let $u_e \in S^2$, $1 \leq e \leq 10$, be the unoriented axes through pairs of opposite icosahedral faces. The zonal degree-six harmonics define

$$L : \mathbb{R}^{10} \longrightarrow \mathcal{H}_6, \qquad L(a)(\omega) = \sum_e a_e P_6(u_e \cdot \omega),$$

where

$$P_6(s) = \frac{1}{16}\left(231s^6 - 315s^4 + 105s^2 - 5\right).$$

The degree-six bond-orientational invariant is the first standard icosahedral channel in the Steinhardt–Nelson–Ronchetti construction [4, 5]. The following theorem identifies its four-dimensional symmetry-breaking component.

Theorem 5.1 (Harmonic Clebsch bridge)**.** `PH-1`*,* `PH-2`*, and* `PH-3`*. Let A be the Petersen adjacency matrix on the ten face axes.*

1. *The map L is injective, and its Gram matrix is*

$$K = \frac{1}{243}(196I + 47J - 112A).$$

Its eigenvalues on $\mathbb{R}^{10} \simeq \mathbf{1} \oplus V_4 \oplus V_5$ are respectively

$$\frac{110}{81}, \qquad \frac{140}{81}, \qquad \frac{28}{81}.$$

2. *Label the axes by the pairs* $\{i,j\} \subset \{1,\dots,5\}$. *The Clebsch summand is the Petersen* (-2)*-eigenspace*

$$a_{ij} = y_i + y_j, \qquad \sum_i y_i = 0.$$

3. *For*

$$F_y(\omega) = \sum_{i<j} (y_i + y_j) P_6(u_{ij} \cdot \omega),$$

the spherical cubic is

$$\frac{1}{4\pi} \int_{S^2} F_y(\omega)^3 \, d\omega = -\frac{784000}{1247103} \sigma_3(y).$$

If $F = \sum_{m=-6}^{6} q_{6m} Y_{6m}$ *in the orthonormal Condon–Shortley convention and*

$$W_6(F) = \sum_{m_1+m_2+m_3=0} \begin{pmatrix} 6 & 6 & 6 \\ m_1 & m_2 & m_3 \end{pmatrix} q_{6m_1} q_{6m_2} q_{6m_3},$$

then

$$\int_{S^2} F^3 \, d\omega = -\frac{130}{\sqrt{3553\pi}} W_6(F).$$

Proof. The face-axis permutation representation decomposes as $\mathbf{1} \oplus V_4 \oplus V_5$, while

$$\mathcal{H}_6|_{A_5} \simeq \mathbf{1} \oplus V_3 \oplus V_4 \oplus V_5.$$

Direct evaluation of $P_6(u_e \cdot u_f)$ gives the displayed Gram matrix. The Petersen eigenvalues $3, 1, -2$, with multiplicities $1, 5, 4$, give the three positive Gram eigenvalues and prove injectivity. For $a_{ij} = y_i + y_j$,

$$(Aa)_{ij} = \sum_{\{k,l\} \cap \{i,j\} = \varnothing} (y_k + y_l) = -2(y_i + y_j),$$

so these vectors form the four-dimensional eigenspace.

The cubic integral is A_5-invariant on V_4. The invariant cubic line is generated by σ_3, so one nonzero value determines the scalar. At $y = (4, -1, -1, -1, -1)$, exact spherical moments give

$$\frac{1}{4\pi} \int F_y^2 = \frac{2800}{351}, \qquad \frac{1}{4\pi} \int F_y^3 = -\frac{15680000}{1247103},$$

while $\sigma_3(y) = 20$. This proves the third displayed scalar.

Finally, the Gaunt formula expresses the triple integral in the orthonormal spherical-harmonic basis as a product of two Wigner $3j$-symbols. Here

$$\begin{pmatrix} 6 & 6 & 6 \\ 0 & 0 & 0 \end{pmatrix} = -\frac{20}{\sqrt{46189}},$$

which gives the stated conversion to W_6. The exact axis, Gram, moment, and Wigner calculations are checked by two independent implementations. $\square$

Order-parameter meaning. A uniform icosahedral neighbour cage lies in the trivial summand. The Clebsch cubic instead sees the V_4 component of nonuniform weights on the ten opposite face pairs. For a face-axis weight vector a, the projector

$$P_{V_4} = \frac{(A-3I)(A-I)}{15}$$

extracts this component. A scale-free signed descriptor is then

$$C_4(a) = \frac{\langle L(P_{V_4}a)^3\rangle}{\langle L(P_{V_4}a)^2\rangle^{3/2}}, \qquad \langle f\rangle = \frac{1}{4\pi}\int_{S^2} f.$$

For the witness in the proof,

$$C_4 = -\frac{120\sqrt{273}}{3553}.$$

Possible weights include opposed-face distortion, chemical decoration, or normal stress. Whether C_4 predicts rearrangement, yielding, or nucleation beyond standard descriptors is an empirical question, recorded as `PH-4`. No applied performance claim is made here.

Remark 5.2 (Which symmetry flips the sign). The cubic is invariant under the rotational A_5-action. The sheet exchange is the odd element in the sign-twisted extension of this action, not an ordinary rotation of a uniformly weighted icosahedron. Any physical realization must therefore specify the decorated face-axis state and the outer-odd operation; the existence of an icosahedral cage alone does not produce the bit.

6 The Klein multiplicity structure

Let X be the Klein cubic threefold and $J(X)$ its intermediate Jacobian. With the standard Tate-twist convention,

$$H^1(J(X),\mathbf{Q}) \simeq H^3(X,\mathbf{Q})(1).$$

Roulleau's period calculation [2] gives an unpolarized complex-abelian-variety isomorphism

$$J(X) \cong E^5, \qquad \mathrm{End}^0_{\mathbf{C}}(E) = \mathbf{Q}(\sqrt{-11}).$$

The ten-dimensional rational representation has no two-dimensional $\mathrm{PSL}_2(11)$- or A_5-stable subspace. Thus the orientation carrier cannot be an equivariant elliptic subvariety. Upon restriction to either icosahedral parent, however,

$$H^1(J(X),\mathbf{Q})|_{A_5} \simeq W_5 \otimes U, \qquad U = \mathrm{Hom}_{A_5}(W_5, H^1(J(X),\mathbf{Q})),$$

where W_5 is the rational irreducible five-space and U is a two-dimensional weight-one Hodge structure.

For the two parents $A_5^{\pm}$, set

$$C_{\pm} = \mathrm{End}_{A_5^{\pm}}(H^1(J(X),\mathbf{Q})).$$

The expected exact relations are

$$C_{\pm} \simeq M_2(\mathbf{Q}), \qquad C_+ \cap C_- \simeq \mathbf{Q}(\sqrt{-11}).$$

The first field records the relative icosahedral position only if it is extracted by a canonical mixed invariant. Arbitrary rank-one idempotents in $M_2(\mathbf{Q})$ are not canonical.

Strong-form target. Using the principal polarization and its Rosati trace form, construct an invariant of the ordered pair (C_+, C_-) whose discriminant class is 5. This would recover $\mathbf{Q}(\sqrt{5})$ transversely to the CM field $\mathbf{Q}(\sqrt{-11})$. The exact rational representation, polarization, commutants, and invariant are assigned to `KL-1`, `KL-2`, and `KL-3`; they remain open under C654.

The Fano surface also contains 55 canonical genus-two curves [3] indexed by the involutions of $\mathrm{PSL}_2(11)$. Their intersection lattice has rank 25 and index 2 in the Néron–Severi group. Whether that saturation detects the finite orientation bit is a second-stage question, not evidence for the strong-form target; it is recorded as `KL-4`.

7 Consequences and boundaries

Once the arithmetic specialization is proved, three consequences become available without creating independent narratives.

1. The cubic-first orientation in the finite matching quotient becomes the reduction of a characteristic-zero quadratic orientation cover.
2. The two golden icosahedral parents become the two fibres of one arithmetic torsor, with the A_4 intersection as their common hinge.
3. The marked Hitchin–Mathieu correspondence becomes a corollary after the two carrier identifications are verified.

Theta, Fourier, quantum, four-sheet holonomy, and degree-23 coherence comparisons are not part of the present theorem complex. They enter only if they can be derived from the orientation cover with precise hypotheses. Otherwise they remain research inventory.

8 Verification and trust

The proof surface distinguishes conceptual arguments, classical inputs, exact certificates, independent replays, and Lean terminals. The machine-readable ledger in `verification/trust_manifest.json` records the current status of every manuscript claim.

The finite tensor bridge uses exact arithmetic in two implementations. The primary route exports the 22 orbit vectors, the sheet signs, the ten-dimensional intertwiner, and the contracted tensor. The independent route reconstructs the group, orbit, quotient, and contraction. Lean checks the final literal 4^3-tensor equality over $\mathbf{F}_{11}$; it does not formalize the upstream orbit construction or all 60 equivariance equations.

The harmonic bridge likewise has a primary exact generator, a canonical JSON certificate, and a separate replay. Both reconstruct the ten face axes over $\mathbf{Q}(\sqrt{5})$, the full Gram and Petersen matrices, and the quadratic and cubic spherical moments. The replay separately checks the Petersen minimal polynomial and the factorial formula for the Wigner $3j$-symbol. Neither route uses floating-point arithmetic.

The arithmetic cover will require a separate certificate bundle: exact golden fibres, comparison matrices, exchanger, spinor class, reduction, and carrier identification, together with an independent replay. The integral model and novelty claims require primary-source proofs rather than a certificate alone. The Klein elevation, if positive, will have its own exact rational matrices and polarization-aware invariant.

The current verification runner checks manuscript structure and trust-ledger coverage only. It intentionally refuses to report release readiness while any load-bearing claim remains gated.

9 Conclusion

The finite Clebsch orientation is already known to be cubic-first: its signed matching tensor restricts to the Clebsch cubic line over $\mathbf{F}_{11}$. The principal remaining problem is to show that this line is the specialization of Hitchin's arithmetic double cover and that its deck exchange is exactly the finite sheet involution.

The strongest form goes further. It asks the Klein intermediate Jacobian to recover the real quadratic orientation field from the relative position of two icosahedral multiplicity structures, while its CM dynamics retain the imaginary quadratic field $\mathbf{Q}(\sqrt{-11})$. These two fields would then have distinct geometric jobs. The commutant calculation decides whether that second statement belongs to the theorem or only motivated its search.

References

[1] N. Hitchin, *Spherical harmonics and the icosahedron*, arXiv:0706.0088.

[2] X. Roulleau, *The Fano surface of the Klein cubic threefold*, arXiv:1001.4853; doi:10.1215/kjm/1248983032.

[3] X. Roulleau, *Genus 2 curve configurations on Fano surfaces*, arXiv:1002.4467; doi:10.14992/00008635.

[4] P. J. Steinhardt, D. R. Nelson, and M. Ronchetti, *Icosahedral bond orientational order in supercooled liquids*, Physical Review Letters **47** (1981), 1297–1300; doi:10.1103/PhysRevLett.47.1297.

[5] P. J. Steinhardt, D. R. Nelson, and M. Ronchetti, *Bond-orientational order in liquids and glasses*, Physical Review B **28** (1983), 784–805; doi:10.1103/PhysRevB.28.784.